Appendix C2 - South-west Corner Marine Park (western arm) Data Analysis Report

South-west Marine Parks Network

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Data Analysis

Statistical methods and model selection results underpinning the natural values reported in Appendix A.

Benthic ecosystem extent

Table C2.1: Habitat model selection results. Top generalised additive models (GAMs) for predicting the probability of occurrence of key habitats from full subset analyses. Differences between the lowest reported corrected Akaike Information Criterion (ΔAICc), AICc weight (ωAICc), variance explained (R^2) and effective degrees of freedom (EDF) are reported for model comparison. Model selection was based on the most parsimonious model (fewest variables - shown in bold) within two units of the lowest AICc. Survey year was offered as a candidate predictor and retained in every top model, so covariate relationships are fitted separately for 2020, 2021, 2024 and 2025.

Response	Model	Δ AICc	ω AICc	R^2	EDF
Macroalgae	**detrended+roughness+Z+year**	**0**	**1**	**0.77664**	**25.78**
Sand	**detrended+roughness+Z+year**	**0**	**1**	**0.6225**	**26.74**
Seagrass	**roughness+aspect+Z+year**	**0**	**1**	**0.51026**	**25.39**
Rock	**detrended+roughness+Z+year**	**0**	**0.768**	**0.5888**	**20.21**
Sessile invertebrates	**detrended+roughness+Z+year**	**0**	**1**	**0.48187**	**25.63**
Reef	**detrended+roughness+Z+year**	**0**	**1**	**0.59209**	**27.15**

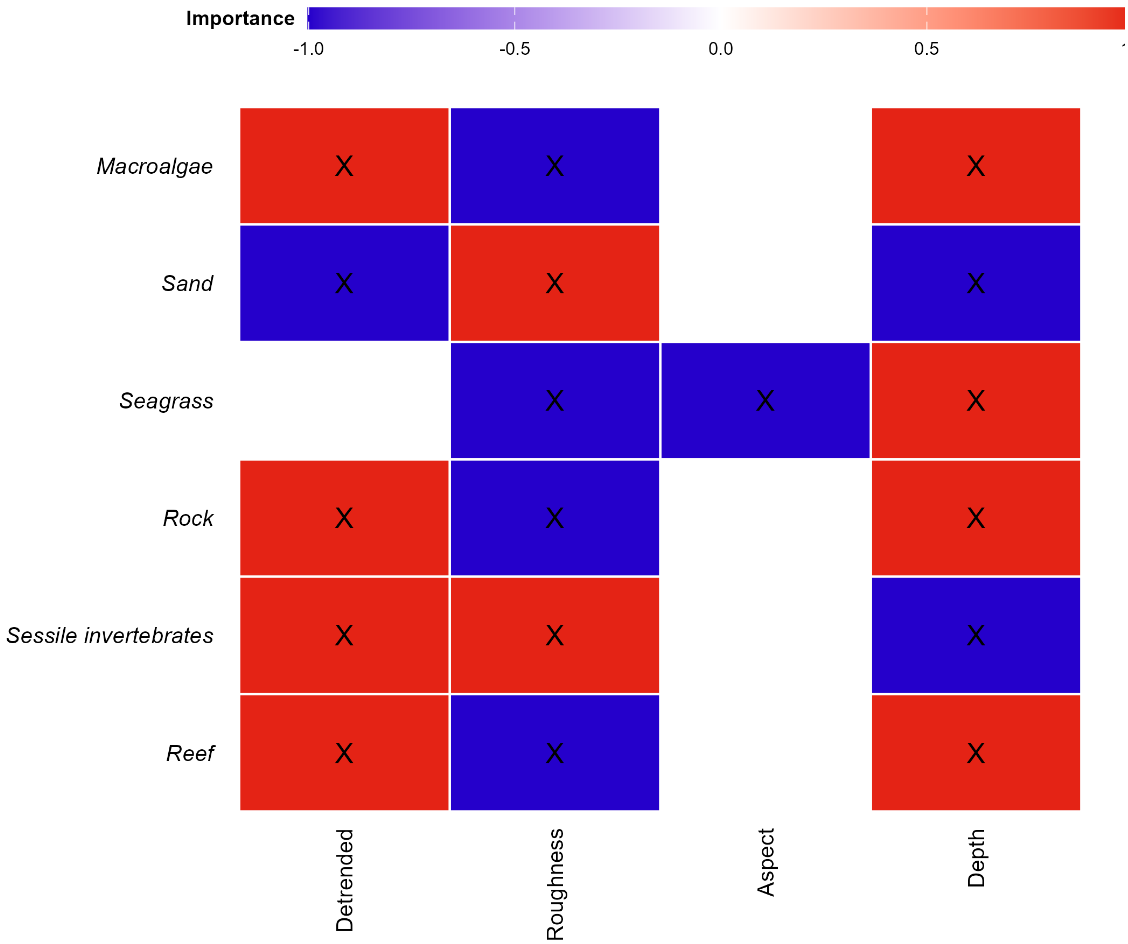

Figure C2.1: Variable importance scores for predicting the probability of occurrence of main habitat categories. Response variables included in the most parsimonious top model are indicated (X), with the colour gradient representing positive (red), zero (white) and negative (blue) relationships. Aspect is a circular variable, so the direction of its relationship should be read from the response curves rather than the colour gradient.

2020

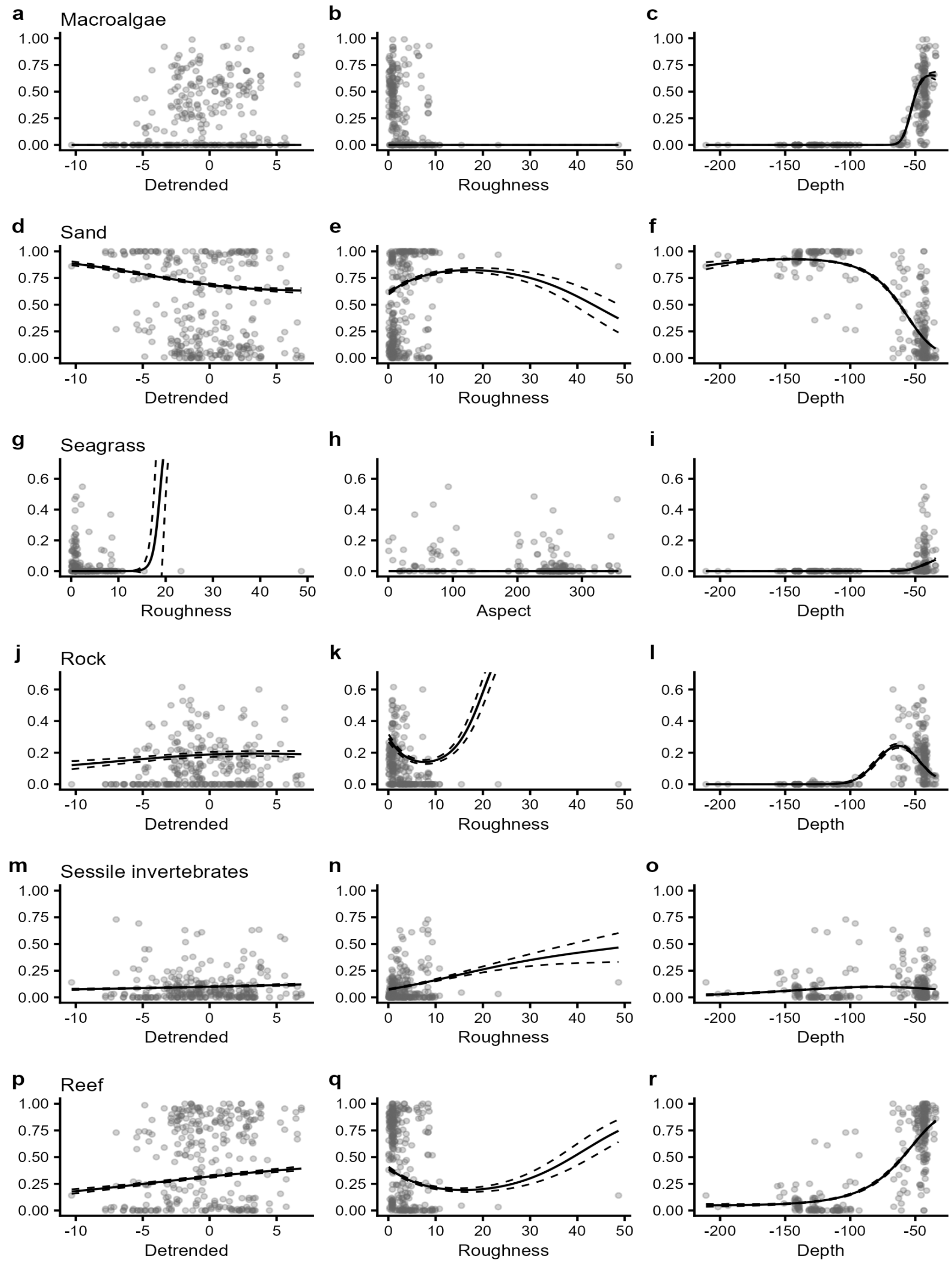

Figure C2.2.1: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2020. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2020. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data for that year.

2021

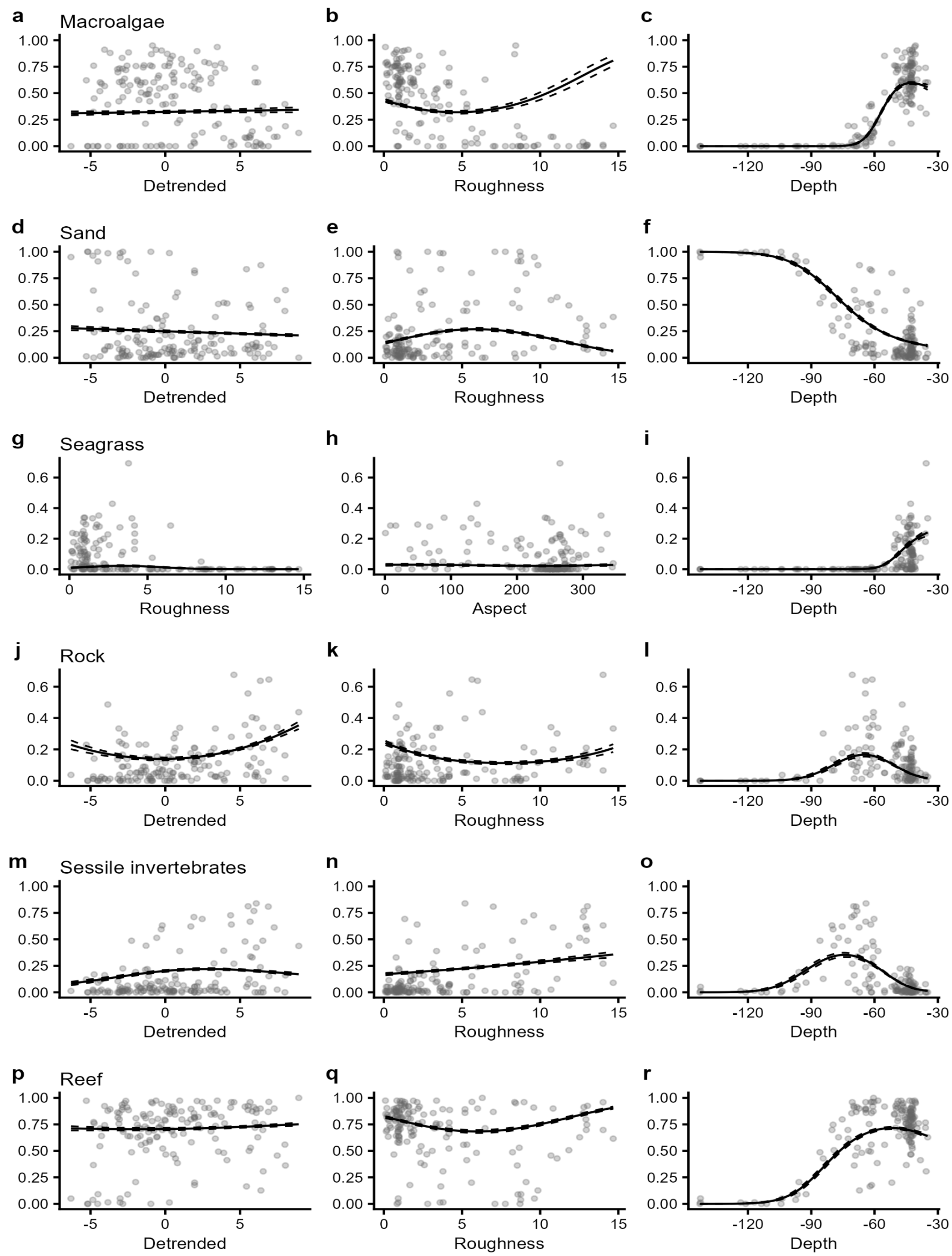

Figure C2.2.2: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2021. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2021. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data for that year.

2024

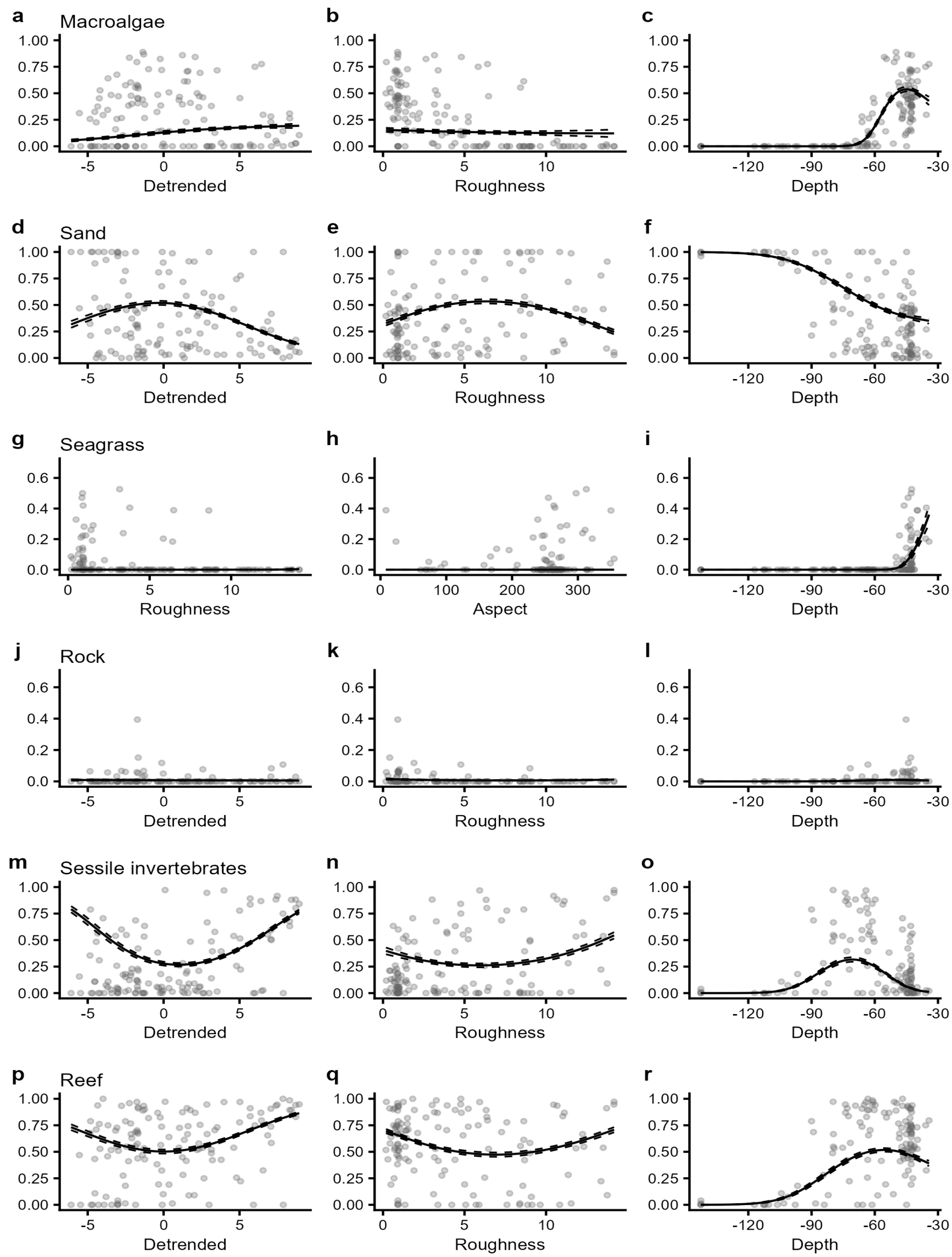

Figure C2.2.3: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2024. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2024. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data for that year.

2025

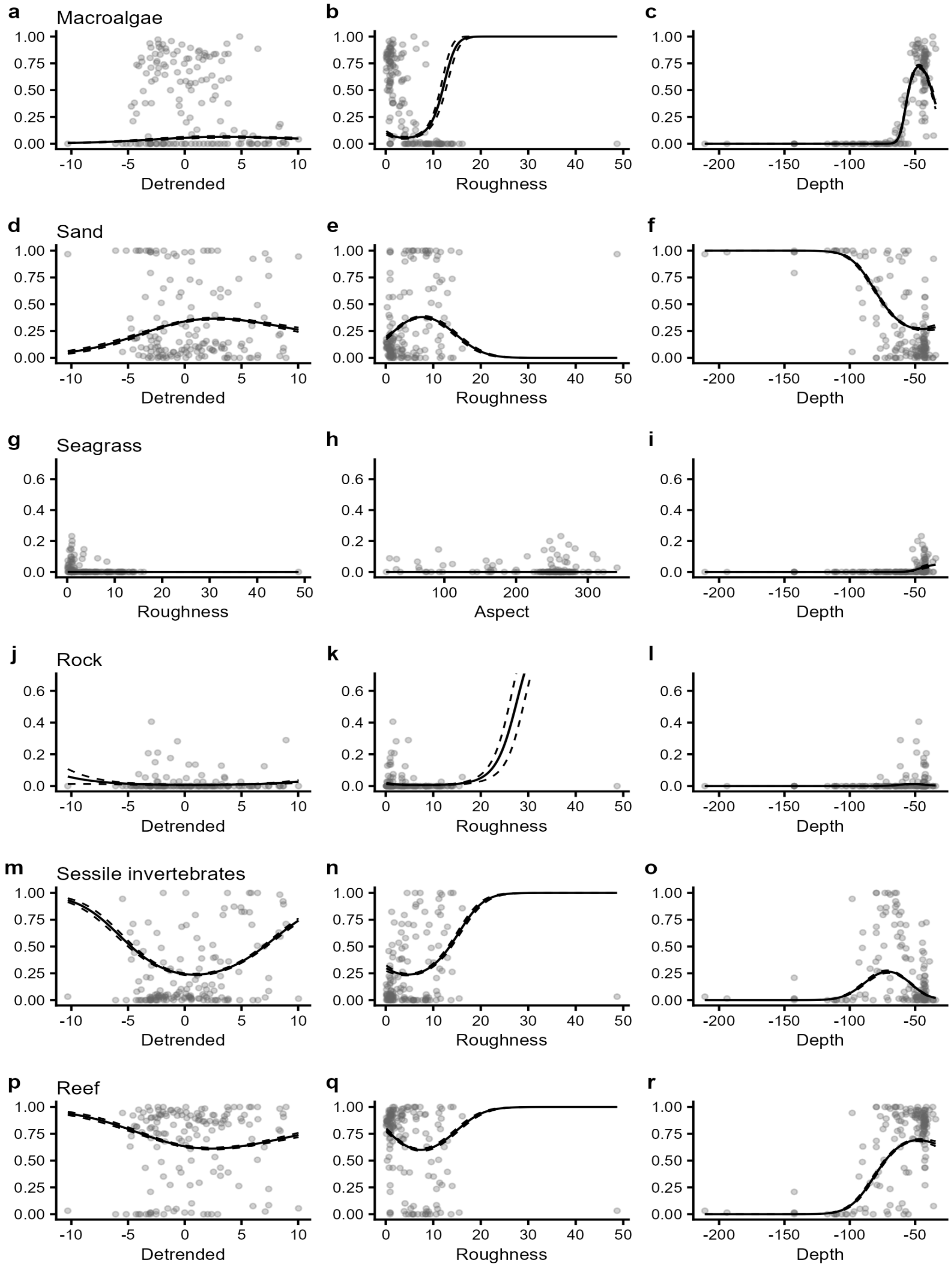

Figure C2.2.4: Plots of the most parsimonious model found to predict the probability of occurrence of key habitat types in 2025. Individual panes display the predicted relationships between model covariates and the probability of occurrence of each habitat class. Solid lines represent predicted Generalized Additive Model (GAM) fits with other variables held constant at their mean value and survey year held at 2025. Dashed lines represent upper and lower standard error bounds around the prediction, and points represent the observed data for that year.